作业 5

1. 彭-陈P. 101, 习题四, 1,2,3,4,5,

2. 证明:

$$\forall \alpha,\beta,\gamma, \frac{\partial b_{\alpha\beta}}{\partial u^{\gamma}} - \frac{\partial b_{\alpha\gamma}}{\partial u^{\beta}} + \Gamma^{\xi}_{\alpha\beta} b_{\xi\gamma} - \Gamma^{\xi}_{\alpha\gamma} b_{\xi\beta} = 0$$

等价于

$$\forall \alpha,\beta,\xi, \frac{\partial b^{\xi}_{\alpha}}{\partial u^{\beta}} - \frac{\partial b^{\xi}_{\beta}}{\partial u^{\alpha}} + b^{\gamma}_{\alpha} \Gamma^{\xi}_{\gamma\beta} - b^{\gamma}_{\beta} \Gamma^{\xi}_{\gamma\alpha} = 0.$$